

1. NB Model

A. Model Structure

$$\begin{aligned}
 Y_i &\sim NB(\mu_i, \alpha) \quad \text{where } i = 1, 2, \dots, s \\
 &\begin{cases} E(Y_i) = \mu_i \\ Var(Y_i) = \mu_i(1 + \alpha\mu_i) \end{cases} \\
 \eta_i &= X_i^t \beta \\
 \log(\mu_i) &= \eta_i
 \end{aligned}$$

B. Likelihood Function

$$\begin{aligned}
 f(Y | \beta, \alpha) &= \prod_i f(y_i | \mu_i, \alpha) \\
 &= \prod_i \binom{y_i + 1/\alpha - 1}{1/\alpha - 1} \left(\frac{1}{1 + \alpha\mu_i} \right)^{1/\alpha} \left(\frac{\alpha\mu_i}{1 + \alpha\mu_i} \right)^{y_i} \\
 &= \prod_i \frac{\Gamma(y_i + 1/\alpha)}{\Gamma(y_i + 1)\Gamma(1/\alpha)} \left(\frac{1}{1 + \alpha\mu_i} \right)^{1/\alpha} \left(1 - \frac{1}{1 + \alpha\mu_i} \right)^{y_i} \\
 &= \prod_{i=1}^n \left[\frac{1}{y_i!} \times \prod_{j=0}^{y_i-1} (j + 1/\alpha) \times (\alpha\mu_i)^{y_i} \times (1 + \alpha\mu_i)^{-(y_i+1/\alpha)} \right] \\
 \\
 l(\mu, \alpha) &= \sum_{i=1}^n \left[\sum_{j=0}^{y_i-1} \log(j + 1/\alpha) - \log y_i! + y_i \log \alpha\mu_i - (y_i + 1/\alpha) \log(1 + \alpha\mu_i) \right] \\
 &= \sum_{i=1}^n \left[\sum_{j=0}^{y_i-1} \log(j + 1/\alpha) - \log y_i! + y_i \log \alpha + y_i x_i^t \beta - (y_i + 1/\alpha) \log(1 + \alpha \exp(x_i^t \beta)) \right] = l(\beta, \alpha)
 \end{aligned}$$

C. Gradient Function

$$\begin{aligned}
 \frac{\partial l(\beta, \alpha)}{\partial \beta} &= \frac{\partial}{\partial \beta} \sum_{i=1}^n [y_i x_i^t \beta - (y_i + 1/\alpha) \log(1 + \alpha \exp(x_i^t \beta))] \\
 &= \sum_{i=1}^n \left(y_i x_i - (y_i + 1/\alpha) \frac{\alpha \exp(x_i^t \beta) x_i}{1 + \alpha \exp(x_i^t \beta)} \right) \\
 &= \sum_{i=1}^n \left(\frac{y_i - \exp(x_i^t \beta)}{1 + \alpha \exp(x_i^t \beta)} \right) x_i^t \\
 &= \sum_{i=1}^n \left(\frac{y_i - \mu_i}{1 + \alpha\mu_i} \right) x_i^t
 \end{aligned}$$

$$\frac{\partial l(\beta, \alpha)}{\partial \alpha} = \sum_{i=1}^n \left[-\alpha^{-2} \sum_{j=0}^{y_i-1} \frac{1}{j + \alpha^{-1}} + \alpha^{-2} \log(1 + \alpha\mu_i) \right]$$

2. ZINB model

A. Model Structure

$$Y_i \sim \begin{cases} 0 & \text{with probability } \psi_i \\ g(y_i) & \text{with probability } 1 - \psi_i \end{cases}$$

$$P(y_i = 0 \mid x_i) = \psi_i + (1 - \psi_i)g(0; \mu_i)$$

$$P(y_i \mid x_i) = (1 - \psi_i)g(y_i; \mu_i), \quad y_i > 0$$

$$\mu_i = g^{-1}(x_i^t \theta)$$

$$\text{logit}(\psi_i) = x_i^t \gamma \quad (\psi_i = \sigma(x_i^t \gamma))$$

$$Y_{ij} = (1 - S_{ij})Z_{ij}$$

$$S_{ij} \sim \text{Bernoulli}(\psi_{ij})$$

$$Z_{ij} \sim \text{NB}(\mu_{ij}, \alpha_j)$$

$$Z_{ij} \perp Z_{ij'}$$

3. CZINB (Consensus ZINB)

- Building Shared Zero model with some similar zero pattern gene sets
- High sparsity gene sets, Moderate sparsity gene sets, Low sparsity gene sets, Almost no sparsity gene sets

A. Model Structure

$$Y_{ij} \sim \begin{cases} 0 & \text{with prob } \psi_i \\ g(y_{ij}) & \text{with prob } 1 - \psi_i \end{cases}$$

$$P(y_{ij} = 0 \mid x_i) = \psi_i + (1 - \psi_i)g(0; \mu_{ij})$$

$$P(y_i \mid x_i) = (1 - \psi_i)g(y_{ij}; \mu_{ij}), \quad y_{ij} > 0$$

$$\mu_{ij} = g^{-1}(x_i^t \beta_j)$$

$$\text{logit}(\psi_i) = x_i^t \gamma \quad (\psi_i = \sigma(x_i^t \gamma))$$

$$Y_{ij} = (1 - S_i)Z_{ij}$$

$$S_i \sim \text{Bernoulli}(\psi_i)$$

$$Z_{ij} \sim \text{NB}(\mu_{ij}, \alpha_j)$$

$$Z_{ij} \perp Z_{ij'}$$

B. Likelihood Function (Core)

Define a cell-level shared-dropout indicator

$$A_i = \mathbf{1}\{Y_{i1} = 0, \dots, Y_{iG} = 0\}.$$

Under CZINB with shared zero probability ψ_i and independent NB latent counts across genes:

$$P(\mathbf{Y}_i \mid x_i) = \begin{cases} \psi_i + (1 - \psi_i) \prod_{j=1}^G f_{\text{NB}}(0; \mu_{ij}, \alpha_j), & A_i = 1, \\ (1 - \psi_i) \prod_{j=1}^G f_{\text{NB}}(y_{ij}; \mu_{ij}, \alpha_j), & A_i = 0. \end{cases}$$

Equivalent log-likelihood contribution for cell i :

$$\ell_i = \begin{cases} \log \left[\psi_i + (1 - \psi_i) \exp \left(\sum_{j=1}^G \log f_{\text{NB}}(0; \mu_{ij}, \alpha_j) \right) \right], & A_i = 1, \\ \log(1 - \psi_i) + \sum_{j=1}^G \log f_{\text{NB}}(y_{ij}; \mu_{ij}, \alpha_j), & A_i = 0. \end{cases}$$

Total log-likelihood:

$$\ell = \sum_{i=1}^n \ell_i.$$

4. MZINB (Marginalized ZINB)

A. Model Structure

$$Y_i \sim \begin{cases} 0 & \text{with probability } \psi_i \\ g(y_i) & \text{with probability } 1 - \psi_i \end{cases}$$

$$\begin{aligned} P(y_i = 0 \mid x_i) &= \psi_i + (1 - \psi_i)g(0; \mu_i) \\ P(y_i \mid x_i) &= (1 - \psi_i)g(y_i; \mu_i), \quad y_i > 0 \end{aligned}$$

$$\begin{aligned} \text{logit}(\psi_i) &= x_i^t \gamma \quad (\psi_i = \sigma(x_i^t \gamma)) \\ \nu_{ij} &= g^{-1}(x_i^t \beta_j) = (1 - \psi_i)\mu_i = E(Y_{ij} \mid X_{ij}) \end{aligned}$$

$$\begin{aligned} \because E(Y_{ij} \mid X_{ij}) &= E(E((1 - S_{ij})Z_{ij} \mid X_{ij})) \\ &= E((1 - \psi_i)Z_{ij} \mid X_{ij}) = (1 - \psi_i)\mu_i \end{aligned}$$

$$\begin{aligned} Y_{ij} &= (1 - S_{ij})Z_{ij} \\ S_{ij} &\sim \text{Bernoulli}(\psi_{ij}) \\ Z_{ij} &\sim \text{NB}(\mu_{ij}, \alpha_j) \end{aligned}$$

5. Test for CZINB

A. LRT ; dropout design without POI

B. Profile LRT ; Fix gamma and separately train the NB model

C. Direct Wald Test

- Hessian : Pseudo-Blockwise matrix. Blockwise Inversion + Sherman-Morrison